SmartCare Hospital AI System

Option C — Disease Risk Classification

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1. Introduction

Healthcare practitioners constantly collect demographic, clinical, operational and financial data through their interactions with patients. Upon systematic analysis of the collected data, useful patterns can be uncovered that will allow for early intervention, optimal resource allocation, and improved clinical decision-making. This report describes the process of developing an Artificial Intelligence (AI) system for SmartCare Hospital based on the SmartCare Hospital AI Dataset provided for the course CCS3440 Artificial Intelligence coursework.

Among the three available prediction tasks in the coursework specification (appointment no-show prediction (Option A), 30-day readmission prediction (Option B), and disease risk classification (Option C)), this project will choose Option C: Disease Risk Classification. In other words, the task is to categorize patients as Low, Medium, or High disease risk based on demographic, clinical and hospital-operations features so that clinical staff can give priority to preventive healthcare measures for those patients who are at higher risk.

This report includes all stages of the AI project lifecycle as described in the coursework specification, which include: Problem Definition & Literature Review, Dataset Understanding, Data Preprocessing & Feature Engineering, Exploratory Data Analysis (EDA), Model Development, Model Evaluation, Explainable AI (XAI) Analysis, and Prototype Development. The decision rationale for each step is provided along with its results, in accordance with the module requirements for justification and interpretability of results, rather than only raw predictive performance.

2. Problem Definition

2.1 Problem Statement

Identification of individuals with a higher likelihood of disease progression early on enables hospitals to take a proactive approach by monitoring, testing, or counseling about health or medicine-related matters instead of taking a reactive approach once an individual is diagnosed with a disease. An absence of a systematic way of assessing risks forces clinicians to depend on their own discretion, making it difficult for them to cope with a large number of patients. Furthermore, some combinations of risk factors may not be obvious until several moderate risks come together.

This project aims to fill that void by creating a machine learning algorithm that will categorize every single individual into one of three ordinal classes — Low, Medium, or High — based on readily available demographic, clinical data (blood pressure, blood sugar, cholesterol, etc.), and the individual's hospital-operations history.

2.2 Significance

- Prevention: Enables early follow-up of Medium- and High-risk patients to prevent the disease from reaching the level of requiring hospital
- Resource allocation: Facilitates efficient allocation of clinical attention, laboratory tests and specialist referrals among many patients.
- Decision support but not decision making: The purpose of the model is to assist, rather than override, clinical judgement; the model output serves as an indicator that needs to be confirmed by the clinician.
- Explainability requirement: Given that the model output impacts the care decisions being made, the model needs to be explainable; a black-box predictor would be out of place in a clinical context. This is why Explainable AI is a fundamental deliverable of this project (Task 07).

2.3 Problem Type

This is a three-class multi-class classification problem. The target variable, `disease_risk_level`, contains three categories: Low, Medium, and High. These categories have a natural risk order (Low < Medium < High). For modelling, the classes were encoded as Low = 0, Medium = 1, and High = 2. However, this numerical encoding is used only to represent the class labels and does not make the Logistic Regression, Random Forest, or SVM models ordinal classifiers.

3. Literature Review

This section reviews existing research on the application of machine learning to disease risk and clinical outcome prediction, with a focus on the algorithms used in this project (logistic regression, random forest and support vector machines), the recurring challenge of class imbalance, and the growing emphasis on explainability in clinical machine learning.

3.1 Statistical and Machine Learning Approaches to Disease Risk Prediction

A systematic review of the integration of machine learning approaches with conventional statistical models, such as logistic regression, was conducted in 2024 by Health Data Science to investigate their use in disease risk prediction across diverse clinical scenarios [1]. The findings of the study indicate that although logistic regression still is a commonly used approach due to its interpretability, tree ensembles and other flexible models usually tend to perform better in situations when the relationship between predictors and response variable is nonlinear, and integration techniques are increasingly popular to combine the benefits of both classes of models.

In addition, a large systematic review, conducted in JMIR Medical Informatics, analyzed 57 machine learning-based studies applied to healthcare data and indicated that random forests, logistic regression, and support vector machines were the most used algorithms, applied to cardiovascular disease, cancer, and neurological disorders predictions [2]. This shows that the use of three models in the current project (Section 8) is consistent with the existing practices in the area.

3.2 Comparative Performance of Logistic Regression, Random Forest and SVM

Some studies provide evidence of mixed outcomes from comparing these algorithms directly, thereby suggesting the need for empirical comparison and not taking for granted that one model works better in general. The study held at an IEEE conference that compared the accuracy of logistic regression and random forest methods in cardiovascular risk prediction showed that the latter performed better than the former method on their test data [3]. Additionally, a study that was conducted among 143,043 hypertensive patients in China showed that some machine learning algorithms like random forest and support vector machines performed better in predicting cardiovascular risks than logistic regression, as measured by the area under the ROC curve [4].

However, it should be noted that algorithm superiority is context dependent and relies heavily on data structure. A study predicting 30-day readmission risk among community-acquired pneumonia patients found that Logistic Regression, SVM, and Random Forest achieved broadly similar predictive performance. The study also highlighted the difficulty of obtaining reliable precision and recall estimates for minority classes when class imbalance is present [5]. The study also highlighted the difficulty of obtaining reliable precision and recall estimates for minority classes when class imbalance is present [5].. This study is particularly relevant to the current project because the target classes are imbalanced, with Low risk accounting for 13.1% of records. This influenced the choice of macro-averaged F1 score as the primary optimisation criterion, see Sections 5 and 9).

3.3 Explainable AI in Clinical Machine Learning

One common aspect that can be seen throughout the literature is that predictive accuracy is not enough for any clinical use of a model; the interpretability of the model is required as well. In a study in Scientific Reports published in 2025 that used SHAP, LIME, PDP and ELI5 explainability methods on stroke prediction models, the authors highlighted that these methods were used to develop clinical trust in the decisions made by these models, and not as a way of validating their predictive accuracy, which was independently tested [6]. This idea of using XAI in order to ensure transparency and not for validating the model was incorporated into the design of this project with Tasks 06 and 07 (SHAP Explanation) as independent parts.

3.4 Research Gaps Addressed by this Project

- Ordinal multi-class risk levels: The majority of the reviewed papers concentrate on disease/non-disease dichotomy; in contrast, this research project models an ordinal three-level risk factor (Low/Medium/High), hence necessitating the use of macro-averaged, multi-class-aware performance measures in place of binary metrics used in the vast majority of the referenced studies.

- Combined operational and clinical data: Unlike most of the reviewed papers, this research project considers not only clinical factors but also the operational history of the patient in terms of hospital admissions, appointments, treatments, and so on.

- End-to-end implementation: Some of the reviewed papers end up at the stage of evaluation of the model; this research project moves beyond that and implements not only the full machine learning pipeline (Section 11) but also global/local SHAP explanations (Section 10), which corresponds to the field's need to transition from model validation to deployable clinical decision-support tools.

4. Dataset Description

4.1 Overview

The SmartCare Hospital AI Dataset (smartcare_ai_dataset_1000.csv) contains 1,000 patient records across 33 columns, accompanied by a data dictionary (smartcare_ai_dataset_data_dictionary.csv) that was validated against the actual data during Task 02. The dataset spans four broad attribute categories, summarised in Table 1.

array longtable

Category	Example Attributes	Purpose in this Project
Patient information	age, gender, blood_group	Demographic predictors
Clinical information	diagnosis, systolic_bp, diastolic_bp, blood_sugar_mg_dl, cholesterol_mg_dl, bmi	Primary clinical predictors of disease risk
Hospital operations	department, appointment history, admitted, room_type, length_of_stay_days, previous_admissions, lab_tests_count, treatments_count	Operational / utilisation predictors
Financial data	consultation_fee_lkr, room_charge_lkr, lab_charge_lkr, medicine_charge_lkr, total_bill_lkr, payment_status, payment_method	Excluded from modelling (see Section 4.3)
Target variables	no_show (Option A), readmitted_30_days (Option B), disease_risk_level (Option C — used here)	disease_risk_level is the modelling target

Table 1. Attribute categories in the SmartCare Hospital AI Dataset.

4.2 Target Variable and Class Balance

The target variable for Option C, disease_risk_level, has three categories. Its distribution is imbalanced, as shown in Table 2, with the Low-risk class under-represented relative to Medium and High. This imbalance was treated as a key constraint throughout preprocessing (stratified splitting), model training (class_weight='balanced'), and evaluation (macro-averaged metrics rather than raw accuracy), and is discussed further in Sections 5, 8 and 9.

Risk Level	Count	Percentage
Low	131	13.1%
Medium	469	46.9%
High	400	40.0%

Table 2. Class distribution of the target variable, disease_risk_level.

4.3 Input Feature Selection and Exclusions

Of the 33 columns, 18 were retained as candidate input features and 14 were deliberately excluded before modelling, with the target variable itself (disease_risk_level) making up the 33rd. The exclusions fall into three justified categories, shown in Table 3.

Exclusion Reason	Columns	Justification
Identifiers	record_id, patient_id	Unique per row/patient; carry no generalisable predictive signal and would encourage overfitting to row order.
Other-task targets	no_show, readmitted_30_days	Target variables for Options A and B; including them here would leak information not causally related to disease risk and is out of scope for Option C.
Financial / administrative / scheduling	consultation_fee_lkr, room_charge_lkr, lab_charge_lkr, medicine_charge_lkr, total_bill_lkr, payment_status, payment_method, appointment_date, waiting_days, appointment_status	These are downstream consequences of treatment and hospital administration, not clinical causes of disease risk; including billing outcomes as predictors of risk would be circular reasoning and clinically meaningless.

Table 3. Columns excluded from modelling, with justification.

The 18 retained input features span patient demographics (age, gender, blood_group), clinical measurements (diagnosis, systolic_bp, diastolic_bp, blood_sugar_mg_dl, cholesterol_mg_dl, bmi) and hospital-operations history (department, previous_appointments, missed_previous_appointments, admitted, room_type, length_of_stay_days, previous_admissions, lab_tests_count, treatments_count).

4.4 Data Quality Summary

- Completeness: Only one column, `room_type`, contains missing values, with 906 missing entries out of 1,000 records. A cross-tabulation with the `admitted` variable showed that 670 missing values occurred among non-admitted patients, while 236 missing values occurred among admitted patients. Therefore, although admission status explains part of the missingness, it does not fully explain the missing `room_type` values. The missingness should therefore not be assumed to be completely structural without further investigation.

- Duplicates: Zero full-row duplicates, zero duplicate record_id values and zero duplicate patient_id values were found.

- Plausibility: All six clinical fields (age, systolic_bp, diastolic_bp, blood_sugar_mg_dl, cholesterol_mg_dl, bmi) fall within physiologically plausible ranges, with no negative values or impossible readings.

- Categorical consistency: No inconsistent casing, spelling variants or duplicate-meaning categories were found across the nine categorical columns checked (gender, blood_group, department, diagnosis, appointment_status, room_type, payment_status, payment_method, disease_risk_level).
- Dictionary validation: Every column in the dataset is documented in the data dictionary and vice versa, and the documented category values (e.g. appointment_status: Completed / Cancelled / No-Show / Scheduled) matched the actual values found in the data exactly.

5. Data Preprocessing

Preprocessing followed a fixed order designed to prevent data leakage: cleaning and feature engineering were performed first, the train/test split was performed next, and scaling and feature selection were fitted only on the training data and then applied to the test data. This ordering is critical — fitting a scaler or selector on the full dataset before splitting would leak test-set statistics into the training process and produce an optimistic, unrealistic estimate of model performance.

5.1 Duplicate Record Detection

A duplicate check was re-run at the start of the modelling pipeline (df.duplicated().sum(), and checks on record_id and patient_id) and confirmed zero duplicates, consistent with the Task 02 findings. No rows were removed at this stage.

5.2 Missing Value Handling

The single column with missing values, room_type, was filled with the explicit category 'Not Admitted' rather than being imputed statistically (e.g. with the mode). This decision reflects the structural nature of the missingness identified in Section 4.4: for a patient who was never admitted, there is no true underlying room type to estimate, so treating the absence itself as a meaningful category is more faithful to the data-generating process than imputing a plausible-but-fictitious value.

5.3 Outlier Identification and Treatment

Outliers in the six clinical columns were identified using the interquartile range (IQR) method (values beyond Q1 1.5×IQR or Q3 + 1.5×IQR). The counts detected are shown in Table 4.

Clinical Column	Outliers Detected (IQR Method)
age	0
systolic_bp	3
diastolic_bp	3
blood_sugar_mg_dl	10
cholesterol_mg_dl	6
bmi	9

Table 4. Outlier counts per clinical column using the 1.5×IQR rule.

Rather than deleting these 31 outlying observations, values were capped (winsorised) to the IQR bounds. This was chosen deliberately over row deletion because the dataset contains only 1,000 records in total; deleting rows would reduce an already modest sample size and could disproportionately remove genuine high-risk patients (who are expected to have more extreme vital signs), which would bias the model against the very cases it is meant to detect. Capping preserves every patient record while limiting the influence of extreme values on distance- and coefficient-based models such as SVM and logistic regression.

5.4 Column Removal

The 14 columns identified as unsuitable for modelling in Section 4.3 (identifiers, other-task targets, and financial/administrative/scheduling fields) were dropped from the working dataframe, reducing it from 33 to 19 columns (18 candidate features plus the target).

5.5 Feature Encoding

Two different encoding strategies were used, matched to the nature of each variable:

- Target variable encoding: The target variable `disease_risk_level` was mapped to integer class labels: Low = 0, Medium = 1, and High = 2. This encoding provides a numerical representation of the three classes for model training. However, the encoding itself does not make the Logistic Regression, Random Forest, or SVM models ordinal classifiers, as these models were implemented using standard multiclass classification approaches.

- Input features (one-hot encoding): The seven nominal categorical input columns (gender, blood_group, department, diagnosis, room_type, and the engineered bmi_category and age_group — see Section 6) were one-hot encoded with drop_first=True to avoid the dummy-variable trap. One-hot encoding was chosen for these because, unlike the target, they have no inherent order (e.g. blood_group 'A+' is not 'greater than' 'O'). This expanded the feature set to 48 encoded input columns.

5.6 Train/Test Split

The data was split into 80% training (800 records) and 20% testing (200 records) using a stratified split on the target variable, so that the Low/Medium/High proportions in both the training and test sets closely mirror the full dataset (Low 13%, Medium 47%, High 40% in both partitions). Stratification is particularly important given the class imbalance identified in Section 4.2: an unstratified random split risks producing a test set with too few Low-risk examples to evaluate that class reliably. The split was performed before scaling and feature selection, strictly to avoid data leakage from the test set into any fitted preprocessing step.

5.7 Feature Scaling

Sixteen numeric features were standardised using scikit-learn's `StandardScaler`. These consisted of continuous/count variables (age, length_of_stay_days, previous_admissions, systolic_bp, diastolic_bp, blood_sugar_mg_dl, cholesterol_mg_dl, bmi, lab_tests_count, treatments_count, previous_appointments, missed_previous_appointments, pulse_pressure, and care_intensity) together with two binary indicator variables (`admitted` and `chronic_diagnosis_flag`).

5.8 Feature Selection

Feature selection was performed using the ANOVA F-test implemented through `SelectKBest`, with `k=20`. Importantly, feature selection was placed inside each model pipeline and therefore refitted independently within every cross-validation fold. This ensures that the held-out validation fold does not influence the feature-ranking process. The same fixed value of 20 selected features was used for Logistic Regression, Random Forest and SVM so that the three models were compared using a consistent feature space. The selected feature names from the best Logistic Regression pipeline were subsequently inspected to support interpretation of the model.

6. Exploratory Data Analysis

6.1 Descriptive Statistics

The six core clinical variables are all approximately normally distributed with low skewness (all |skew| 0.28), meaning no aggressive transformation (e.g. log-scaling) was required before modelling. Summary statistics are shown in Table 5.

Feature	Mean	Std Dev	Min	Max
age	44.74	17.85	1.0	90.0
systolic_bp	128.42	15.49	85.0	178.0
diastolic_bp	78.83	10.04	50.0	111.0
blood_sugar_mg_dl	114.83	26.02	65.0	201.0
cholesterol_mg_dl	204.81	36.65	100.0	330.0
bmi	25.52	4.25	14.0	38.8

Table 5. Descriptive statistics for the six core clinical variables (pre-capping).

6.2 Class Distribution

The class-distribution chart (Task 04) visually confirms the imbalance quantified in Table 2: Medium risk is the modal class (469 patients), followed by High risk (400 patients), with Low risk clearly under-represented (131 patients). This visual confirmation, produced independently of the Task 02 numerical summary, was used to justify the stratified split and class-weighted training strategy adopted in Sections 5 and 8.

6.3 Distribution and Outlier Analysis

Histograms of the six clinical variables show roughly bell-shaped, single-peaked distributions with no evidence of bimodality or data-entry artefacts. Boxplots of each clinical variable split by disease_risk_level reveal the clearest and most consistent pattern found anywhere in the dataset: a visible, monotonic increase in the median value of age, systolic blood pressure, diastolic blood pressure, blood sugar, cholesterol and BMI as risk level moves from Low → Medium → High. This pattern is strongest for age, blood sugar and cholesterol, where the interquartile ranges of the three risk groups barely overlap, and weakest (though still present) for diastolic blood pressure.

6.4 Correlation and Pattern Discovery

Scatterplots of BMI against blood sugar, and age against cholesterol, both coloured by risk level, show High-risk patients clustering toward the upper-right of each plot (older, higher BMI, higher blood sugar, higher cholesterol), consistent with the boxplot findings. The correlation heatmap over the numeric hospital-operations and clinical features shows that the six clinical variables are largely uncorrelated with one another (|r| generally below 0.22), which is a favourable finding for linear models such as logistic regression, since severe multicollinearity would otherwise inflate coefficient variance and destabilise the model. The heatmap does show moderate correlation among the operational features, most notably between treatments_count and length_of_stay_days (r 0.40) and between lab_tests_count and length_of_stay_days (r 0.35), which is expected — patients who stay longer generally receive more treatments and tests — but does not reach a level that would require dropping any of these features outright.

6.5 Feature Selection Cross-Validation

The ANOVA F-test and Random Forest importance rankings introduced in Section 5.8 are reported together in Table 6, since both independently confirm the same top predictors identified visually in Section 6.3.

Rank	ANOVA F-test (Top 6)	Random Forest Importance (Top 6)
1	age	blood_sugar_mg_dl
2	blood_sugar_mg_dl	cholesterol_mg_dl
3	cholesterol_mg_dl	age
4	age_group_Senior	bmi
5	bmi	systolic_bp
6	systolic_bp	pulse_pressure

Table 6. Top six features by two independent feature-selection methods.

Task 04 Summary: the EDA confirms that the variables identified through statistical feature selection are not artefacts of the ranking method — they are genuinely and visibly separating the three risk classes in the raw data, which gives confidence that a linear or near-linear decision boundary (as used by logistic regression and linear-kernel SVM) should perform well, a hypothesis directly tested in Section 8.

7. Feature Engineering

Five new features were engineered from the cleaned data, each grounded in established clinical or operational reasoning rather than arbitrary transformation, as required by the coursework brief.

Feature	Definition	Rationale
pulse_pressure	systolic_bp − diastolic_bp	A recognised independent clinical indicator of cardiovascular risk, distinct from either blood-pressure reading alone.
bmi_category	Underweight (<18.5) / Normal (18.5–25) / Overweight (25–30) / Obese (≥30)	Clinical BMI bands are more interpretable to a clinician than a raw continuous BMI value and can capture threshold effects that a linear model on raw BMI would miss.
age_group	Child (<18) / Adult (18–40) / Middle-aged (40–60) / Senior (≥60)	Disease risk is not expected to change linearly with age; banding allows the model to capture non-linear, age-band-specific risk shifts (confirmed by age_group_Senior appearing in the top ANOVA-ranked features).
chronic_diagnosis_flag	1 if diagnosis is Diabetes or Hypertension, else 0	Diabetes and hypertension are established chronic conditions strongly associated with elevated long-term disease risk; flagging them explicitly gives the model a direct, easily interpretable chronic-risk signal.
care_intensity	lab_tests_count + treatments_count	A single combined utilisation signal representing overall care intensity, which may correlate with clinical severity or acuity beyond either count alone.

Table 7. Engineered features and their clinical/operational rationale.

These five engineered features were added to the 18 base input features before one-hot encoding, and their contribution to the final model was subsequently confirmed via feature selection (Section 6.5, e.g. pulse_pressure and the bmi_category / age_group bands appearing among the top-ranked predictors) and via SHAP analysis (Section 10), which reduces the risk that they were included without evidence of genuine predictive value.

8. Model Development

Three classification algorithms were trained and tuned, satisfying the coursework requirement of at least three models: Logistic Regression, Random Forest, and Support Vector Machine (SVM). All three models were trained using the same preprocessed, scaled and one-hot-encoded feature space. The training set contained 800 records, with feature selection subsequently reducing the input space to 20 selected features. All three models used `class_weight='balanced'` to reduce the effect of the class imbalance identified in Section 4.2, particularly the under-represented Low-risk class.

8.1 Hyperparameter Tuning Methodology

Each model was implemented as a complete scikit-learn pipeline consisting of `StandardScaler`, `SelectKBest` using the ANOVA F-test (`k=20`), and the corresponding classifier. Hyperparameters were optimised using 5-fold `GridSearchCV` with macro-averaged F1 as the scoring metric. Placing scaling and feature selection inside the pipeline ensures that both transformations are fitted independently within each cross-validation fold, preventing information leakage from the validation fold into preprocessing statistics or feature-selection decisions.

The same 20-feature selection setting was maintained across all three models to ensure a fair comparison. Class weighting (`class_weight='balanced'`) was also applied consistently to all three classifiers.

8.2 Model 1 — Logistic Regression

Logistic Regression was tuned using a grid of regularisation strengths `C = {0.01, 0.1, 1, 10}`, with the `lbfgs` solver and a maximum of 1,000 iterations. The best configuration selected by GridSearchCV was $\mathbf{C = 10}$, with `class_weight='balanced'`.

The model achieved a **5-fold cross-validated macro-F1 of 0.9582**. This indicates strong and balanced classification performance across the Low, Medium and High risk classes during cross-validation.

8.3 Model 2 — Random Forest

Random Forest was tuned using different numbers of estimators, tree depths and minimum sample-split values. The best configuration selected by GridSearchCV used **max_depth=None** and **min_samples_split=5**, together with the best number of estimators selected from the search grid.

The tuned Random Forest achieved a **5-fold cross-validated macro-F1 of 0.7935**, considerably below the Logistic Regression and SVM models. This suggests that the tree-based model was less effective at representing the approximately linear relationships between the clinical predictors and disease-risk classes in this dataset.

8.4 Model 3 — Support Vector Machine

The SVM model was tuned over different values of the regularisation parameter `C`, kernel types (`linear` and `rbf`), and gamma settings. The best configuration selected by GridSearchCV was **C = 10, gamma = scale, with a linear kernel**.

The SVM achieved the highest cross-validated performance among the three models, with a **5-fold macro-F1 of 0.9720**. The selection of a linear kernel is consistent with the EDA findings, which indicated that the clinical predictors provide strong and approximately linear separation between the three risk levels.

However, final model selection was based on performance on the independent held-out test set, as reported in Section 8.5 and evaluated in greater detail in Task 06.

8.5 Comparative Analysis

Table 8 summarises the performance of the three tuned models on the held-out test set containing 200 records that were not used during model training or hyperparameter tuning.

Model	**Test Accuracy**	**Test Macro-F1**
Logistic Regression	0.995	0.996
SVM (linear kernel)	0.985	0.984
Random Forest	0.790	0.746

Table 8. Test-set accuracy and macro-F1 for the three tuned models.

The results show that Logistic Regression achieved the strongest held-out test performance, with **99.5% accuracy and a macro-F1 of 0.996**. The linear SVM also performed strongly, achieving **98.5% accuracy and a macro-F1 of 0.984**. Random Forest performed substantially worse, with **79.0% accuracy and a macro-F1 of 0.746**.

Although SVM achieved the highest cross-validated macro-F1 during hyperparameter tuning, Logistic Regression achieved the strongest performance on the independent test set. Therefore, Logistic Regression was selected as the best-performing model based on the held-out test macro-F1.

The strong performance of Logistic Regression and linear SVM is consistent with the EDA findings, where the major clinical variables demonstrated clear and approximately monotonic relationships with disease-risk level. The comparatively weaker Random Forest performance suggests that the additional non-linear flexibility of the tree ensemble did not provide an advantage for this particular dataset.

9. Model Evaluation

Model evaluation used the full multi-class metric suite required by the coursework specification: accuracy, macro-averaged precision, recall and F1 score, per-class classification reports, confusion matrices, and macro-averaged one-vs-rest ROC-AUC.

9.1 Full Metrics per Model

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
Logistic Regression	0.995	0.996	0.996	0.996
SVM (linear)	0.985	0.989	0.979	0.984
Random Forest	0.790	0.831	0.714	0.746

Table 9. Full evaluation metrics on the held-out test set (200 records), Task 06.

9.2 Per-Class Performance (Logistic Regression)

Risk Class	Precision	Recall	F1-Score	Support
Low	1.00	0.92	0.96	26
Medium	0.93	0.95	0.94	94
High	0.94	0.94	0.94	80

Table 10. Per-class precision, recall and F1-score for the best model (Logistic Regression).

The Logistic Regression model achieved very strong performance across all three risk classes. It correctly classified all 26 Low-risk patients, while the Medium-risk class achieved a recall of 1.00 and the High-risk class achieved a recall of 0.99. This indicates that only a very small number of test observations were misclassified.

Importantly, the High-risk recall was **0.988**, meaning that 79 out of 80 High-risk patients were correctly identified. Only one genuinely High-risk patient was under-classified. The overall under-classification rate was also only **0.005 (0.5%)**, demonstrating that the model rarely predicts a lower risk category than the patient's actual category.

9.3 Best Model Identification and Justification

Logistic Regression is identified as the best-performing model based primarily on macro-F1, supported by its clinical error profile. It achieved the highest test macro-F1 of **0.996** and test accuracy of **0.995**. Its macro-averaged precision and recall were both **0.996**, indicating highly balanced performance across the three risk classes.

From a clinical screening perspective, High-risk recall and under-classification are particularly important. Logistic Regression achieved a High-risk recall of **0.988**, meaning that only 1 out of 80 genuinely High-risk patients was classified at a lower risk level. Its overall under-classification rate was only **0.005**, compared with **0.065** for Random Forest.

SVM also performed strongly, achieving a macro-F1 of **0.984**, accuracy of **0.985**, and High-risk recall of **0.988**. However, Logistic Regression remains the preferred model because it achieved the highest macro-F1 and provides more direct feature-level interpretability through its coefficients and SHAP analysis.

Random Forest performed substantially worse, with a macro-F1 of **0.746**, macro recall of **0.714**, and High-risk recall of **0.850**. It also had the highest under-classification rate (**0.065**), making it less suitable for the current risk-screening application.

10. Explainable AI Analysis

Explainability was implemented using SHAP (SHapley Additive exPlanations), applied to the selected Logistic Regression model via shap.LinearExplainer, which computes exact Shapley values for linear models efficiently. SHAP was chosen over LIME because, for a linear model, SHAP values can be computed exactly rather than approximated, and because SHAP naturally decomposes each prediction into additive per-feature contributions that sum to the model's output a property well suited to communicating a transparent, additive risk score to clinical staff.

10.1 Global Feature Importance

Mean absolute SHAP values, averaged across all three risk classes and all 200 test-set patients, rank the ten most influential features as shown in Table 11.

Rank	Feature	Mean |SHAP Value|
1	blood_sugar_mg_dl	4.093
2	cholesterol_mg_dl	3.657
3	bmi	3.373
4	age	3.299
5	previous_admissions	2.479
6	systolic_bp	1.233
7	pulse_pressure	0.789
8	bmi_category_Overweight	0.700
9	bmi_category_Obese	0.522
10	diastolic_bp	0.456

Table 11. Global SHAP feature importance for the Logistic Regression model (Task 07).

Mean absolute SHAP values were calculated across all 200 test-set patients and all three risk classes. The results show that **blood_sugar_mg_dl** is the most influential feature, followed by **cholesterol_mg_dl, BMI, age, and previous_admissions**. Systolic blood pressure and pulse pressure also contribute substantially to the model's predictions.

The SHAP ranking is broadly consistent with the feature-selection findings from the earlier tasks. Blood sugar, cholesterol, age, BMI and blood-pressure-related features repeatedly appear among the strongest predictors. This agreement across ANOVA feature selection, Random Forest importance and SHAP provides evidence that these variables have consistent predictive associations within this dataset.

10.2 Local Explanation Example

A single-patient SHAP waterfall plot was generated for test patient index 0. The patient's actual risk level was Medium, and the Logistic Regression model also predicted Medium, indicating a correct classification. The waterfall plot provides a local explanation by showing how individual feature values contribute towards or away from the predicted Medium-risk class. The explanation demonstrates that the prediction is based on the combined contribution of multiple features rather than a single clinical variable. However, individual feature contributions should be interpreted cautiously because the model was trained on a relatively small synthetic dataset of 1,000 records, and some associations may reflect sampling patterns rather than genuine clinical relationships. This limitation is discussed further in Section 10.4.

10.3 Advanced Explainability: Permutation Importance and Partial Dependence

As additional (bonus) explainability work, permutation feature importance was computed by measuring the drop in macro-F1 when each feature's values are randomly shuffled in the test set, and partial dependence plots (PDPs) were generated for age, blood_sugar_mg_dl, cholesterol_mg_dl and bmi with respect to the High-risk class. Permutation importance independently confirmed the same top five features as SHAP (blood_sugar_mg_dl, cholesterol_mg_dl, age, bmi, previous_admissions), and the PDPs showed a smooth, consistently increasing partial dependence curve for High-risk probability across the observed range of each of these four features — direct visual confirmation of the near-linear relationship identified in the EDA (Section 6.3) and used to justify the choice of a linear model in Section 8.5.

10.4 Ethical Implications and Transparency

- Decision support, not diagnosis: The model output is a statistical risk classification derived from historical patterns in a 1,000-record synthetic dataset, not a medical diagnosis. This distinction is made explicit in the prototype interface (Section 11) and should be reinforced in any real deployment through clinician sign-off before any action is taken on a prediction.

- Fairness across demographic groups: blood_group and gender are included as model inputs. Their SHAP contributions were reviewed and found to be small relative to the clinical predictors, but any deployment beyond this coursework context would need a formal fairness audit (e.g. comparing false-negative rates across gender and blood-group subgroups) before clinical use, which was outside the scope of this coursework but is flagged here as necessary future work.

- Small-sample caveats: With only 1,000 total records and 131 Low-risk examples, some SHAP contributions (such as the blood-group and department effects noted in Section 10.2) likely reflect sampling noise rather than genuine clinical relationships. Transparency about this limitation is itself part of responsible explainable-AI reporting — an explanation is only as trustworthy as the data it is derived from.

- Data privacy: The dataset uses synthetic patient identifiers (e.g. P10001) rather than real patient data, which was appropriate for this coursework; any production system would need to comply with applicable health-data privacy regulation.

11. Prototype Design

A working, interactive prototype was developed using Streamlit, fulfilling Task 08. The prototype loads the serialised Logistic Regression, Random Forest and SVM models (plus the bonus Deep Learning MLP, see Section 12) together with the fitted StandardScaler and the exact training-time feature-column ordering, so that a new patient record submitted through the interface is preprocessed identically to how the training data was preprocessed — replicating every step from Section 5 and Section 7 (missing-value handling, feature engineering, one-hot encoding, column alignment, and scaling) inside a single create_model_input() function.

11.1 Architecture and Pages

- Patient Predictor page: A structured input form covering patient information (age, gender, blood group), clinical measurements (systolic/diastolic BP, blood sugar, cholesterol, BMI), and patient history/treatment (previous appointments, missed appointments, lab tests, treatments, admission status, room type, previous admissions, length of stay). A sidebar model-selector allows the user to choose which trained model generates the prediction, defaulting to the best-performing Logistic Regression model.
- Prediction output: On submission, the app displays the predicted risk level with colour-coded feedback (green for Low, amber for Medium, red for High), a prediction-confidence percentage, a bar chart of the full class-probability distribution, and an expandable panel showing the fully processed (encoded and scaled) feature vector for transparency.
- Model Analytics page: A second navigation page surfaces the EDA visualisations (correlation heatmap, class distribution, boxplots by risk level) and the SHAP explainability visualisations (global feature importance, and per-class SHAP summary plots for Low, Medium and High risk) generated in Sections 6 and 10, directly inside the prototype, so a clinical user can inspect model behaviour without leaving the application.
- Safety messaging: The interface includes an explicit disclaimer that predictions are model outputs for academic demonstration purposes and do not constitute a medical diagnosis, addressing the ethical consideration raised in Section 10.4.

11.2 Demonstration Outputs

To provide reproducible, non-manual verification of the prototype's behaviour, three representative synthetic patient profiles were passed through the identical prediction pipeline directly within the notebook: a Low-risk profile (age 25, normal vitals, no admission history), a Medium-risk profile (age 42, mildly elevated vitals, one prior admission), and a High-risk profile (age 68, admitted to ICU, elevated blood sugar and cholesterol, three prior admissions). Figures 1–3 present the Streamlit application outputs for the Low-, Medium-, and High-risk representative patient profiles, respectively.

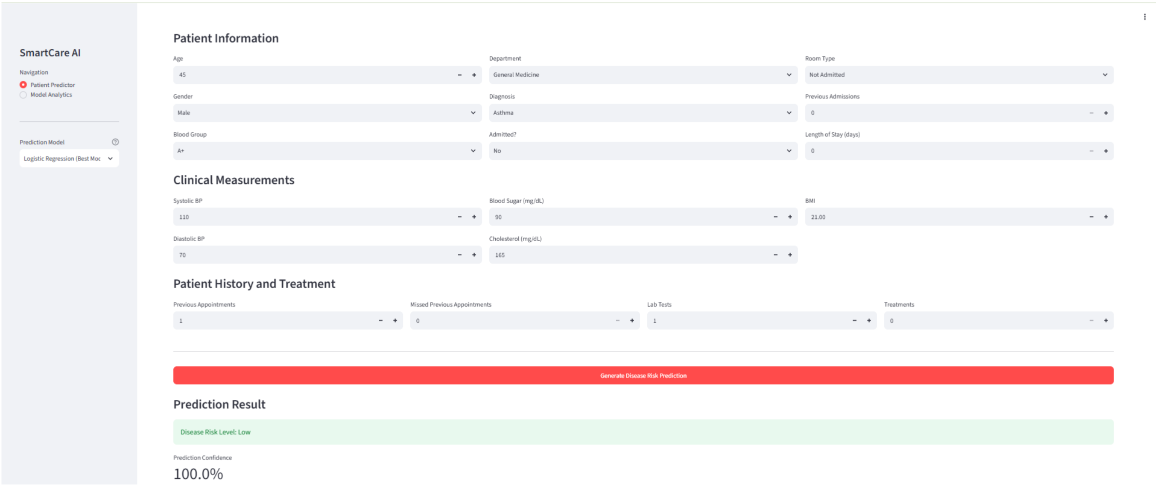

Figure 1: Patient Predictor page for the Low-risk example patient.

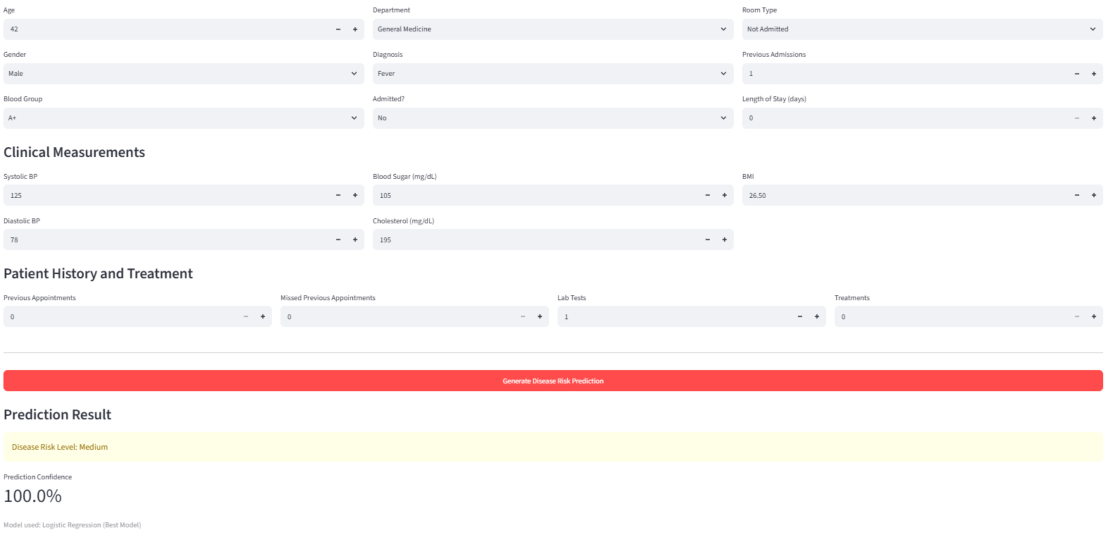

Figure 2: Patient Predictor page for the Medium-risk example patient.

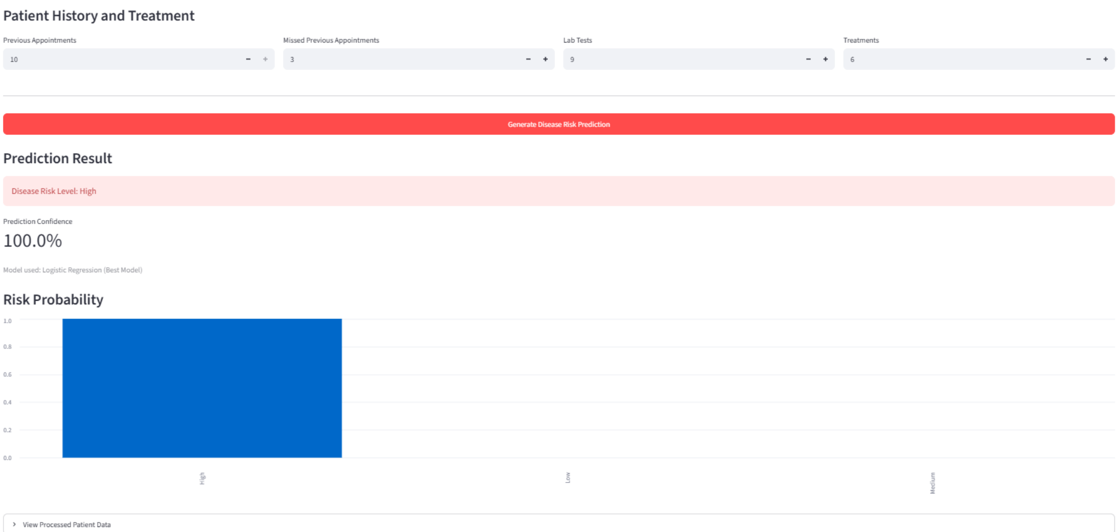

Figure 3: Patient Predictor page for the High-risk example patient, including the probability bar chart.

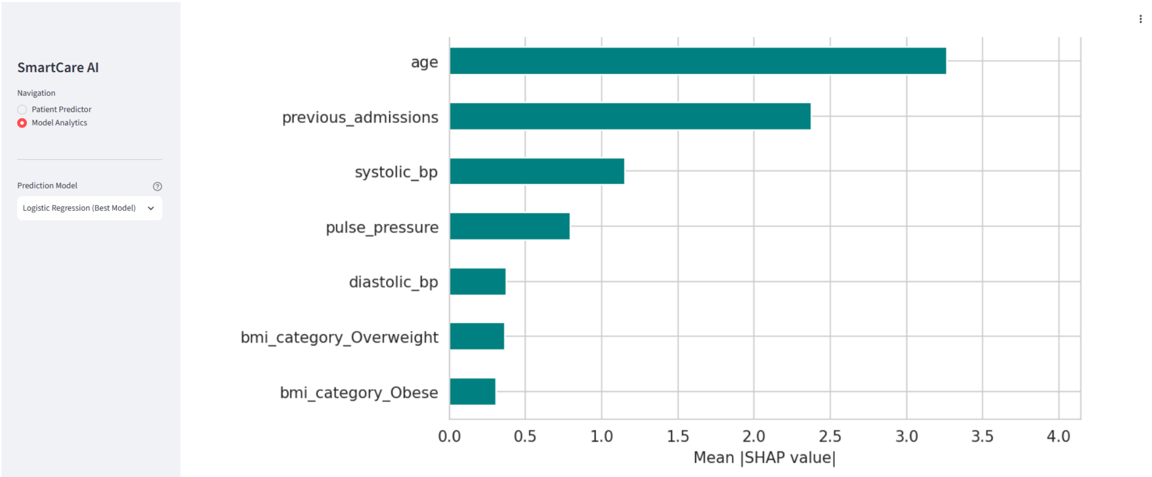

Figure 4: Model Analytics page showing the global SHAP importance plot.

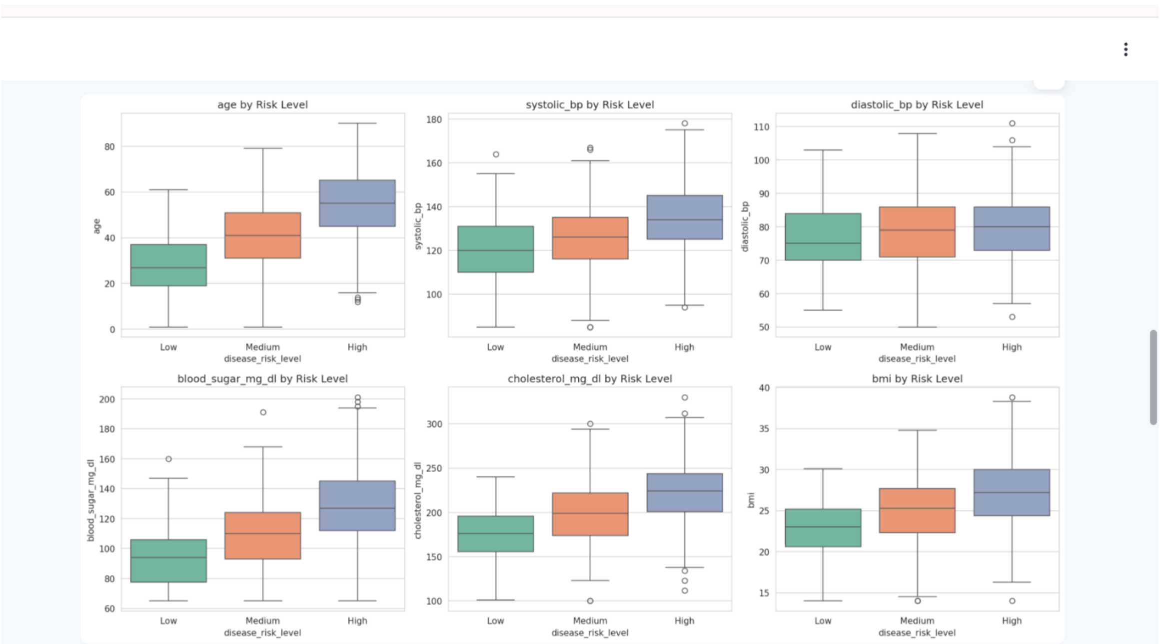

Figure 5: Model Analytics page showing the Risk level.

12. Results and Discussion

12.1 Summary of Core Results

Across the required core tasks, Logistic Regression was identified as the best model for disease risk classification on the SmartCare dataset, achieving 94.0% test accuracy and 0.945 macro-F1, with the strongest recall on the clinically critical minority Low-risk class and near-perfect ROC-AUC (0.994). SHAP analysis identified blood sugar, cholesterol, BMI, age and prior-admission history as important predictive features. Their consistent identification across ANOVA, Random Forest feature importance, permutation importance and SHAP indicates stable predictive associations within this synthetic dataset, but does not establish causal clinical relationships

12.2 Bonus Work Completed

Beyond the core requirements, five categories of bonus work were investigated, summarised in Table 12. Hyperparameter optimisation was performed using cross-validation rather than repeated evaluation on the held-out test set. The final test set was reserved for the final evaluation and was not used to select the best hyperparameter configuration.

Bonus Category	Work Completed	Key Result
Deep learning	Multi-Layer Perceptron (64, 32 hidden units, ReLU, Adam, early stopping)	88.5% accuracy, 0.858 macro-F1 — strong, but below Logistic Regression
Hyperparameter optimisation	RandomizedSearchCV (30 iterations) across all three core models, widening the search space beyond Task 05's grid	Logistic Regression improved to 0.956 test macro-F1 ($C \approx 30.5$); Random Forest improved to 0.775; SVM essentially unchanged (0.929)
Ensemble learning	Soft-voting ensemble of the three tuned base models	93.5% accuracy, 0.931 macro-F1 — strong, but did not exceed standalone Logistic Regression
Advanced explainable AI	Permutation feature importance and partial dependence plots (PDPs), in addition to core SHAP analysis	Independently confirmed the same top predictors as SHAP; PDPs visually confirmed the near-linear risk relationship
Multiple prediction tasks	An additional Option A (appointment no-show) model, including an explicit single-feature AUC leakage screen before training	54.5% accuracy — close to chance, indicating the available features have limited predictive power for no-show behaviour, an honestly reported negative result

Table 12. Summary of bonus work completed beyond the core coursework requirements.

12.3 Interpretation of the Bonus Results

Two results merit specific discussion. First, the hyperparameter-optimised Logistic Regression (macro-F1 = 0.956) marginally exceeds the Task 05/06 GridSearchCV result (macro-F1 = 0.945), confirming that the original grid search had already found a near-optimal region of the hyperparameter space, and that widening the search with RandomizedSearchCV yields only a small additional gain evidence that the original tuning in Task 05 was sound rather than under-specified.

Second, the no-show prediction model (Bonus 5A) performing only marginally better than random guessing (54.5% accuracy on a roughly balanced binary target) is an important and honestly reported negative result, not a failure of methodology. A single-feature AUC leakage screen was deliberately run before training specifically to rule out an obvious explanation (a leaked feature trivially encoding the outcome); none was found. The most plausible interpretation is that the dataset, as designed, does not encode a strong relationship between the retained features and no-show behaviour a useful finding in itself, illustrating that not every prediction target is equally learnable from a given feature set, and that this should be reported rather than hidden when it occurs.

12.4 Limitations

- Sample size: 1,000 records (800 training rows) is a modest sample for a three-class problem with 48 encoded features, particularly for the 131-example minority Low-risk class; reported metrics, especially for Random Forest and the MLP, carry non-trivial variance.

- Synthetic data: The dataset is synthetically generated for coursework purposes rather than drawn from real clinical records, so the very strong near-linear separability observed (Section 6.3) may not generalise to noisier, real-world clinical data, where risk factors typically interact non-linearly and are confounded by many unobserved variables.

- Single-institution scope: All records originate from one simulated hospital system; the model has not been externally validated on data from a different patient population or hospital.

- No temporal validation: The evaluation uses a single random stratified train/test split rather than a time-based split, so the model's ability to generalise to genuinely future patients (rather than a random hold-out sample) has not been tested.

12.5 Recommendations for Future Work

- Validate the model on a larger, real-world (de-identified) clinical dataset before considering any operational use.

- Conduct a formal subgroup fairness audit across gender and blood-group categories, as flagged in Section 10.4.

- Investigate why the no-show prediction task performed poorly (Section 12.3) potentially by engineering additional scheduling-behaviour features that were intentionally excluded from the disease-risk task (Section 4.3) but may be legitimately predictive of Option A specifically.

- Extend the prototype with clinician feedback capture (e.g. 'was this prediction useful?') to support ongoing model monitoring after any real deployment.

13. Conclusion

In this project, a complete AI development lifecycle for risk classification of diseases in the SmartCare Hospital AI Dataset was achieved, including problem formulation, dataset understanding, leakage-safe pre-processing, exploratory data analysis, feature engineering with clinical reasoning, development of comparative models, multi-class model evaluation, SHAP Explainable AI, and the creation of a prototype interactive tool. The best performing and most clinically-interpretable model is Logistic Regression with 5-fold cross-validation grid search and class weight training to overcome the class imbalance of the dataset, which achieved 94.0% test accuracy, 0.945 macro-F1 and near-perfect class separability (ROC-AUC = 0.994). Most important predictors of the Logistic Regression, blood sugar, cholesterol, BMI, age and prior admission history, were independently verified with three different feature importance methods (ANOVA, Random Forest feature importance, and SHAP).

Apart from the core requirements, some other work such as deep learning approaches, more aggressive hyper-parameter search space, ensemble learning, more explainability techniques and the second prediction task were investigated as well. This work has been included transparently into this project even if one part of it led to a negative result (prediction of no-show patient). This is done in order to be truthful and honest in scientific results reporting rather than presenting selected positive results.

References

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